

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0745

River Street 138 kV
March 30, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0745. The Interconnection Customer proposes a 288.2 MW wind facility interconnecting at the River Street 138 kV substation in the Pennsylvania area. The request was received on February 14, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 222 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$59,753,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lindsay to W Marinette Sw Yd 138kV	Riv St to Vitro Architectural Glass Pa 138kV	150.7	180	100.0
River St to Lindsay 138kV	Vitro Architectural Glass Pa to Riv St 138kV	139.3	199	100.0
Riv Street to Vitro Architectural Glass Pa 138kV L	Maunabo 4301 to West Marinette Sw Yd 138kV	126.5	220	68.9
W Marinette Sw Yd to Maunabo 4301 138kV	Vitro Architectural Glass Pa to River St 138kV Line	111.6	244	57.2
Point Meadows to Maunabo 4301 138kV	Vitro Architectural Glass Pa to River St 138kV	104.1	254	57.2
Haskell Ave to Seney Uppco Substation 138kV	Vitro Architectural Glass Pa to River Street 138kV	101.7	201	7.6
Susquehanna to Pt Meadows 138kV	Riv Street to Vitro Architectural Glass Pa 138kV	100.4	258	57.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lindsay to W Marinette Sw Yd 138kV	Rebuild line with higher-capacity conductor	\$12,155,000	100% (\$12,155,000)
River St to Lindsay 138kV	Rebuild line with higher-capacity conductor	\$13,570,000	100% (\$13,570,000)
Riv Street to Vitro Architectural Glass Pa 138kV L	Rebuild line with higher-capacity conductor	\$11,413,000	100% (\$11,413,000)
West Marinette Sw Yd to Maunabo 4301 138kV	Rebuild line with higher-capacity conductor	\$4,859,000	100% (\$4,859,000)
Point Meadows to Maunabo 4301 138kV	Reconductor line	\$1,275,000	100% (\$1,275,000)
Haskell Ave to Seney Uppco Substation 138kV	Reconductor line	\$14,136,000	100% (\$14,136,000)
Susquehanna to Pt Meadows 138kV	Reconductor line	\$2,345,000	100% (\$2,345,000)
	Total allocated to project		\$59,753,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.